

ActInf ModelStream 012.1 ~ Fisher, Whyte, and Hohwy: An Active Inference Model of the Optimism Bias

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all right hello and welcome it is June 27th Z UTC 2024 and we are in active inference model stream number 12.1 with Beth fiser and yaka howi so we will have a presentation and a discussion on an active inference model of the optimism bias so thank you both for joining and Beth to you for the presentation great thanks Daniel um yep so I'm Beth and I'm going to talk about work we've just put out as a pre-print and active inference model of the optimism bias and this is work we've done along with Chris White who I think a lot of you will already know or read some of his work and yeah we're here in Australia at the Mones Center for Consciousness and contemplative studies and this is part of my PhD project so I'm doing my PhD with yako so what I'll do is I'll first of all give a bit of of an overview of what the optimism bias is and why this is interesting to study why I'm interested in it I'm then going to talk a bit about um the field of computational Psychiatry so I'm using active inference not so much in a way to like to understand the nitty R of active inference but more as an application as a tool in computational Psychiatry so I thought it would be helpful to explain how we can use these models in in these kinds of settings and and the purpose of why I'm using these models then I'll go through the simulations uh that we have in the paper and then just some conclusions so the optimism bias is a cognitive bias where our expectations are better than reality and what's interesting about the optimism bias is that you might think in order to make the best decisions in life and have the best life outcomes you should represent information as it is in reality and not have expectations that are better than reality but the optimism bias shows us that if we think things are better than they are and overestimate the likelihood of good things happening we have all these better life outcomes so people who are optimistic have a lower risk of all CA ality and reduce risk of cardiovascular disease they have decreased fatigue and chronic illness and they engage in more self-care behaviors there's even this really cool study that showed that people who are optimistic had improv cognitive functioning following

a traumatic brain injury um optimists they just earn more money they have better immune systems and they just have less anxiety and worry um overall so why would it be that being optimistic and thinking things are better than they are leads to all these positive things so to explain that I'm going to go through an example so imagine here you're someone who holds the reality beliefs so in reality it's true that 50% of marriages end in divorce if you work out you reduce your risk of heart disease by 14% And if you eat well you reduce your risk of cancer by 10% but then imagine that you're someone who's optimistic so if you're optimistic you might think no only 10% of marriages and in divorce if I uh work out I'll reduce my risk of heart disease by 40% and if I eat well I'll reduce my risk of cancer by 30% so if we hold these optimistic beliefs we're going to be way more likely to do these things so we'll be more likely to eat well we're going to be more likely to work out we're going to be more likely to go out there with the hope of finding a partner and although not everything will work out for you because the reality statistics are there on the left by going out and engaging in more activities and things in life we we increase our chances of some things working up it's like the idea of minimizing the risk of missed opportunity if I do go out there and work out and go to parties and eat well and all these things I'm going to have more likely chance of some of them working out than if I just stay at home and don't do anything so being optimistic increases our engagement in the world and this overall is a positive thing that can lead to all these health benefits so if optimism is really good for all these things what is the connection between optimism and mental health so people with an anxiety and depression have been shown to lack optimism they can either have expectations that worse than reality so this can be pessimism or they can have expectations that match reality so this is sometimes called depressive realism which as we've kind of gone over that this isn't a good thing either so what I'm interested in is well could we increase optimism as an intervention for depression and anxiety so that's what what I what I what I want to do so to explain how I'm going to do this and and how active inference fits in into this goal uh I'm just going to explain a bit about computational Psychiatry and and what we're kind of aiming to do in in that space and I'll go through the steps of of what you want to do for your aim so the first thing which I guess in in most areas in science you have to come up with a a conceptual idea so to do this for this project I read a lot about active inference I read a lot about optimism and try to think how active inference could best model optimism what predictions we could maybe make from these models and and how we can model that so you want to spend a lot of time on this first step coming up with a good conceptual idea and I'm sure uh a lot of you who have done the active inference work you

know that there there's a lot to that that step that's an important step but once you come up with your idea uh you can run these simulations so that's what we've done in this paper so thought a lot about optimism how it fits in active inference and then ran some things uh on my computer to see okay well can I simulate what I want to see am I seeing optimism come out in active inference in the way that I want it to and then the next step after that the idea is that after you've run your simulations you kind of have a good base to then fit the data so what's nice about running the simulations is you kind of have more background knowledge before you go out and just start start fitting and collecting data so you can design your experiments to to fit what you found in in the simulations um my image here is of these rats because we have other work where we've done um active inference data fitting to to a r model so that's what's going on there and then the point of all of this is that hopefully in the end we can come up with a model that will have an application in the clinic so after we've done our simulation and our data fitting well the idea to use the optimism as the example we'll have some model that then clinicians can use to help measure and understand how optimism is um working for a patient and then they can use that model to to design treatments for that person so that's the whole goal and aim of what I'm doing and that's why I'm doing what I want to do and that's why and how I'm using active inference so in terms of the optimism plan this paper is talking about the simulation step and then hopefully from there we can do some data fitting for this work so then there are just some other things I wanted to say before we start talking about the models about when we're thinking about models to use in the clinic there's some considerations um that are important that might be different to considerations if you're using active inference and modeling in other set but if we want to um use them in computational Psychiatry we actually want simple models because if the end goal is that a psychiatrist or psychologist or or or doctor uses them we don't want anything super complicated because um that won't be practical that won't be that won't be used so one of the things when I'm designing my models is I'm trying to to use models that that are simple that can give us more information about it ass sort um but is still can still be used and then the idea is that this can help with Precision Psychiatry so at the moment um you can go to the doctor you do a self report for depression you get diagnosed with depression but there isn't really much understanding of the mechanisms that are involved of of of what caused the depression or different things that may have caused it so hopefully the these models that we design can um we can understand okay well maybe that person is experiencing these uh depressive symptoms because they have low optimism or maybe they don't have low optimism maybe there's something else that that's

going on um that we can understand through the use of these models so I hope that kind of sets the scene of of this work and um and and hopefully that made sense so when I did all the reading of the literature and on active inference and optimism we this is our conceptual idea of an active inference model of the optimism bias that is the Precision over the positive outcomes for the agent um and that might not sound that groundbreaking given the definitions of optimism I've given uh but we'll go through the models to see how we can fit this in into the framework so what we did is that we formalize optimism as a state factor in the generative model that then could be applied to different simulations and tasks so we have this um I'm not sure if you can see the mouse very easily but we have this oh we can um optimism State Factor here and we have an optimistic State and a pessimistic State and the idea is that you can take this state factor in different situations and different task settings and we wanted to do something like this because optimism is this you know this higher higher level belief that affects different areas of life it's it's not just uh constrained to one one setting so we wanted to think okay well how can we use something that can be General and applied to different different tasks and can we um use this in the simulations and the optimism State Factor controls or it's conditions the Precision of the updating for the positive outcomes and we'll go through the examples and hopefully that will start making more sense how that works so when I go through the rest of the talk I'll refer to optimism bias as levels and I just wanted to explain what I mean when I talk about that so we refer to these different optimism levels depending on the optimism State Factor so here we would say this is optimism level 1 um because there it's 0.1 optimism 0.9 pessimism you will have 0.5 optimism and then this would be fully a level one optimism sorry um so when I go through the rest of the presentation this is what when I'm referring to levels this this is what I mean so we ran well we three different simulations but we refer to them as experiments in this paper so the first one was we simulated the loss of the optimism bias so we wanted so one of the things um that's interesting about optimism is it Sean across species so we see it um in Bay in horses all animals have optimism and it's this this is it's an Adaptive adaptive bias and it suggests that the optimism bias is something that we're born with it's something it's not like other biases that that we learn later on it's something that is is a part of us and there's a lot of research showing that optimism um over if depending on your environment in childhood you can have lower levels of optimism so factors that you're exposed to early on can influence the level of optimism that you have and we wanted to kind of do this first experiment with this developmental stage to simulate how optimism might be lost or maintained during development so that's the idea of the first experiment the second

experiment uh we applied our optimism by a state factor that was lost or or maintained in experiment one to a belief updating task that's used in the literature on optimism so this is used in human participants uh to study optimism and we wanted to do this because hopefully this can ground our model in actually empirical literature and then if we're able to replicate results that is found in the empirical literature and optimism it kind of gives our model we we kind of have more hope that this is a model that we can then go out and use so that was the idea of that step and then the third step was we ran a two Arm Bandit task um but it's it's a modified two Arm Bandit task where we wanted to see if we could measure optimistic action so um because optimists engage in the world more we wanted to see if we can develop a task that can measure that kind of Engagement so I'll go through the details of that a bit when we get to that step so the first uh simulation I'll go over is the simulation of the loss of the optimism bias so as I've already kind of mentioned there are environmental factors that affect optimism level so this can be um you know things that happen to you in childhood if you're exposed to trauma um your socioeconomic status these kind of things so people who usually have gone through more adversity during childhood have low optimism and another thing is that people who are optimistic have improve cardiovascular responses to stress so this means that they have a healthier arousal outcome so if they get really stressed out or exposed to something they they don't have this the super high um AR which which as we know is isn't very good for you and then on the other hand as probably most of you know anxiety is related to high arousal so if you're anxious um you can have just higher arousal overall and also when there's there's nothing there to really warrant that that arousal so what we wanted to do was to simulate this loss of optimism bias over time by exposing the agents to higher and lower rousel and positive and negative violence events and um the idea behind that is that obviously we know that there's uh higher arousal you can have high arousal from positive and negative things and you can have low arousal from positive and negative things and a healthy agent would be able to have all of those equally and and and that's that's a good thing whereas some someone who is anxious or they can sometimes experience this higher arousal um as a negative thing if even when it's not so this is what we wanted to see if we expose the agents to these things during development does this change their optimism and we also looked at how it looked uh affected their arousal mappings but we'll go through that in the figure so for this experiment so this is just our little image of the experiment so we have the agent who's exposed to all of different combinations of these events so positive high and low negative high and low and for this experiment we have a hierarchical active inference model so this means that we have two

levels so the first level of this model just set up the vigilance arousal States so we just fed the model different um vigilance and arousal States and there was no learning there was no no actions it was just just feeding those observations that the that the agent had so this is our first level and then on our second level this is the agents optimism so this is happening at a slower time time course they're getting these uh vigilance and arousal observations um over these these weeks that that's how we operationalize it and this is changing affecting their their optimism so in the hierarchical models um the the states of of level one they give the observations for level two so whatever state the agent was in in this level one it gives them their their observations there so on this second level the agent was learning two things so they were learning their their optimism which is um our state Factor here and they were also learning their arousal likelihood Matrix so over time um if they're in an optimistic or pessimistic State the mapping between high and low arousal so they're the two things that that we were looking at so our first result so this is what the optimism looked at after the simulation of development and we we ran it for 200 a so this is optimism this is optimism State Factor remember we talk about those levels and then the proportion of negatively valenced events they had so what we found is that agents that were more optimistic after development were exposed to less negative vean events which is what we would hope to see and that makes sense um agents that were less optimistic were Expos opposed to more negative violence events also something we would expect to see but what we can also see in this plot is there is quite a lot of variability uh but this is actually a really good thing because when we're thinking about um humans they have a lot of variability so it's it's nice to have a model that that can show this so there's also instances um let's say an example this so This agent here is pretty optimistic but they were EXP to lots of negatively valenced events so this can be the things if someone well the agents if they're initially exposed to a lot of good things earlier on it can kind of be protective so when they get these negative Veen events later on it doesn't change their model as much it doesn't change their optimism as much so we can think of that as protective or we have agents here who weren't exposed to many negatively valenced events um but they don't have much optimism and this could be because maybe they're exposed to these events much earlier on in the in the simulation um and this yeah so this is also with what what we would see we can also see in in the literature is that good things that happen can be protective against the more negative things that can happen so this with this was called to say that they had the variability so then this so that was the optimism and now we're looking at the likelihood matrices from the simulation so this is the likelihood matricies

that look at the um optimism and pessimism States versus the low and the high arousal so here we have an agent who has optimism level 07 so this is a pretty pessimistic agent and what we can see is that the higher r outcomes there's a lot there's a lot higher Precision between P the pessimistic State and the higher Ral outcomes so what this means is that these agents get higher risal outcomes and it's like okay I'm in a pessimistic state I'm in a pessimistic state so that this this isn't this is um what we would expect to see with someone who who is anxious so this High arousal it's getting giving a lot of information that you're in in a certain State and then here we have this optimistic agent who has a completely flat likelihood Matrix and this is a healthy healthy U mapping because the whether you're experiencing high or low arousal doesn't um give you any you can be in either State and that's fine um so that's again how we would think of of healthy individuals uh they can experience High arousal and that can be from a good or a bad thing and they it doesn't give them a signal or something's bad's happening right away so this was also we were excited to see this um this kind of mapping in the likelihood Matrix from the simulation we also found a lot of variability in these as well these are just two examples because we ran 200 different agents um but again variability for us and what we're doing is a good thing so this just gives us um yeah more ways we can look at at different individuals so that we were happy to see the variability as well so just to sum up experiment one um we found that exposure to more negative veence events reduced optimism which is what we wanted to see we found that pessimistic agents Le higher Precision for higher arousal which inferred pessimistic state and then we found optimistic agents had a flat likelihood Matrix so Rouser did not influence their state so these were two things that that we wanted to find so now I'll go on to experiment two so for experiment two we use the belief updating task that's used in literature so this is a standard optimism task that's used used in in humans and the way this task works is is someone a participant is asked to estimate the likelihood of a bad thing happening so what's the likelihood um you'll develop cancer and you give that estimation and then the participant is shown the actual likelihood of of that estimation and that likelihood can be lower or higher than their estimation so if they're told their risk of of something bad is lower that's considered a good news trial because they've just been given good news whereas if they're told their risk of something is worse that's a bad news trial because they've just been given bad news and you can look at how much participants update to good news so how much do they change their beliefs when they get good news versus how much they change their beliefs when they get bad news so it's measur it's measured here so we can see that this is the control so controls are are optimistic because

people are generally optimistic and we can see that they update more to good news than bad news and then we have the um participants who have major depressive disorder and they don't have this same asymmetry when they when they update to good and bad news so we wanted to use this task uh to see if the optimism State factor that we simulated or in this first experiment could then be used in the belief updating task um to to show if we can see the results that we find in the literature because if we can then show that uh well agents who are more optimistic by by our model how we're defining optimism update uh have this same asymmetry and belief updating and that pessimistic agents don't then our model it's it's kind of more evidence to show that that our model might be capturing something that is that's that's useful so I'll go over the model here so we set up the task so we had the optimism and pessimism State Factor so if a if the agent was in an optimistic state it updated more to good to a good outcome than a bad outcome if the agent was in a pessimistic state it updated more to a bad outcome s a good outcome so this is this the um likelihood Matrix here the there was a initial hidden State factor which was the belief about the good versus bad outcome so the agent gets gets this belief about the good versus bad outcome they're then fed observations which is either um better than expected or worse so good news or bad news so they give good news or bad news observations and then so then they are updating their uh hidden State Factor belief from the observation of good versus bad news and whether they're in an optimistic or pessimistic State this influences how much they update to good news so the optimism State factor is conditioned on on this updating here and then we took the difference between the initial hidden State Factor belief so that's here compared to the hidden State Factor belief after the observation so that's the D in then in the next step and we took the difference to look at the uh good versus bad news for the updating for this we actually ended up running so usually in in the task that human participants run they do 70 trials so we wanted to do 7 trials the way that this best worked though was to actually run 70 different models so we so each model is um each trial is a different model and then we take the results of of each of those um which yeah I mean I guess that can we can get into a discussion about modeling and what we're modeling and how it's working but I suppose you can think of like well if we have a model model of my belief about car accidents and we have a model about my belief about cancer they could all be different models in for the for this purpose it's not it's it's a different belief that we're we're it's a different concept I guess um I hope this is all making sense it's I haven't done an online talk for ages and it's and it's um hard to know how this this is gone Okay so we've got our belief updating task and let's look at the results so here we have our optimism

State Factor so this is the optimism level so this is a very optimistic agent this is a very not optimistic agent and we find that the optimistic agent updates more to good versus bad news so we can see that here um we can see that the more pessimistic agents update more to bad versus good news and then here we have this figure from the study that looked at optimism in controls versus people with major depression uh depressive disorder and we have this here to show that that we did kind of replicate this this um pattern in in belief updating in our simulation so that was exciting so the major so if if these if the people with MDD uh have more real depressive realism than they would be this optimism level⁵ and we can see yeah that we that we are replicating this result um so that was that was exciting because this we were able to show so we found that our belief updating task simulation results were comparable to the literature and optimism um so we say that this is kind of some evidence for our optimism model because we're able to replicate things that that are already out there and this also provides a good basis for a model that can be tested for fitting data so if you think about the first when I was talking about the the goals of of using models in the clinic we want to have models that we can also collect data and and fit them for and not all models um you could be able to do that so easily it can be it can be tricky so this is a model that we can then we can now just go and get people to do this task and fit the model to see if if we then get the same um optimism State factor and we can also look at their their likelihood matrices so that's stuff that we can do all right so now I'm on to the last experiment so we have simulated the loss of optimism or maintained it depending on on the development of the agent we've shown how it's um worked in this belief updating task and now we're going to look at optimistic action in this two arm banded task and this in this two Arm Bandit task uh what we did was the agent could either select left or right um so left was highrisk high reward right was low risk low reward or there was a third option where the agent could select stay so this was like the safe option so if you selected stay you didn't lose any money but you didn't win any money and this is an important uh thing to be out of model for what we're interested in because if if if you are depressed and you are not optimistic one of the things that you can tend to do is stay at home and not go out and engage in all those things that that I was talking about at the start of the talk so it's it's helpful if we can see um to model this what what who would be decide making this decision to stay stay you know just stay safe stay at home not but not getting any rewards overall and one of the uh this is kind of related to optimism but one of the um interventions for depression at the moment is behavioral activation in cognitive behavioral therapy which is the idea to get people to go out in in the world more and engage in more

things so it would be good if we could have a task that we can measure this decision making so that's what we did so in this task yeah we have our three actions that so this model involved policies so uh we've had a hierarchical model to begin with with we had this second model had no policies and then this is where the agent this is where our optimistic action comes in and here again we have our optimism versus pessimism State Factor so this is what's been going on through out and so here we have the generative process which is actually just the structure of the two Arm Bandit task so if I select left um we've got 70% chance of a large win 30% chance of a large loss if I select right we've got 60% chance of a small win 40% chance of a small loss so this is what's going on in the actual task and this is the belief of the agent so if you're optimistic you have this High Precision over the large win and the small win so just wins in general and if you're pessimistic you have this large Precision over the loss over the losses rather than the wins and this is controlled by this State Factor here this is how the agents updating their beliefs um and we ran I think it was 60 trials for for this task so here are our results for this task the total winnings and so again we've got our optimism level so this would be an agent with optimism level .9 optimism level .1 so this is pessimistic agent and we can see that the more optimistic an agent is the more winnings they have overall so in this task setting um having this Precision over these positive outcomes was led to this these higher winnings the the agent who had the Precision over low outcomes actually ended up losing money but was also important about this figure is that we can see that it wasn't that the most optimistic agent had the most winnings we see that the agent here has has the most winnings and a important idea when we're thinking about optimism and interventions is it probably isn't going to be the case that 100% optimism all the time in every situation is a good thing different environments that we're in there's going to be different levels of optimism that that are going to be most beneficial and that's another thing that's what's so nice about um being able to do these models is that we can kind of explore what those different levels look like and um and yeah test that out in different environments rather than just saying oh optimism is good let's make everyone optimistic without actually really understanding how it's going on in this sense so findings like this I are are important so then we wanted to look at well what are what's the actions of these agents that led them to this these different outcomes so we have our pessimistic agent here and we can see that they just selected left and right and then they just like no that's it I'm just going to not I'm just going to stay so we can think of this as somebody who might be depressed who goes out has some bad times and they're like no I'm just going to

stay at home now that's that's that's not going to give me any any of Rewards or anything that I'm looking for uh we have this middle agent who kind of went back and forth between between the left and right arm and then here we have the optimistic agent who went right left and then just stayed in the left arm which is the high risk High reward but overall that would that would lead you to the best um the mo the most money in the end uh so this this is nice because we can see um what kind of actions uh the actions of how of what kind of actions would result from the different belief belief that these agents have and this is the behavior that we've been talking about so our two Arm Bandit task uh was able to capture optimistic action and we could see that the optimism promoted the engagement action because the optimistic agents didn't select stay they they kept going in the task selecting left and right we also found that full optimism is not the most beneficial in this task setting which is an important consideration especially if we're thinking about interventions oh and then yep so they were the three simulations that we ran so with this with these three different simulations we feel like this is pretty good uh initial support for an active inference model of optimism there's a lot of uh positives about modeling optimism with this active inference model we can use it in these different task settings we can we've set up models that we can now collect data for um so that's the first conclusion our model show demonstrated how optimism could be lost in development which is um an important consideration uh belief updating task grounded the model in literature and optimism so we're able to show okay this is how it fits to things that are already out there measuring optimism and our two unbanded task was able to show how optimistic action um was able to measure the action so the next step is to uh fit participant data to the optimism model and and we've designed these models with that in mind um so a lot of the choices that we made were for this next step uh which is going to be an exciting step I think to see if we can um yeah get see if this this is we can get data to fit and and then hopefully use these models as a tool in the clinic thank you awesome awesome um great well should I keep sharing this um if somebody ask something we can bring it back up but um great thank you for the for the presentation um yob do you want to give any initial thoughts or or no uh this is uh a result of Beth's brain power and so I'll leave word to to Beth on this um okay so a few questions I wrote down and then also um we'll see what else people ask so what what brought you from the review of how people had modeled optimism to this relationship with anxiety and arousal so what so sorry so how why was I interested in the optimism and the arousal yeah okay because it seems like you could be very chill and have a a high believe that you know the lottery is going to be better than expected or something like that so what what brought you

into to looking at the relationship with arousal specifically yeah um that's a really good question it was thinking about events that happen in childhood and development um that can C like so so what happens when when a child has a traumatic event what happens when they're in an uncertain environment um so initially that was the idea and then we're thinking about this arousal seems to play a part in in these traumatic events and that is when um when children are developing they are learning this this kind of mapping so we first think of that and then we looked at the literature on optimism and arousal and then it's like oh actually optimism actually helps maintain uh healthy arousal so it was kind of it was just a search I guess and finding thinking about what happens during development that might affect optimism what's the nature of those experiences so what is it about those experience that cause a child to feel a certain way or have a certain belief obviously there's something going on in in the body those things colle uh connected to anxiety and then that yeah and then it happened to be like oh this is this is cool and interesting now because we're reading on optimism and arousal and optimism is good at maintaining that can we fit those things together uh so we kind of explored it in that way and I guess what's also nice about running the simulations is when we come up with those ideas uh when maybe not so they're interesting ideas we're not 100% sold on them and then you run the simulations and then you're like oh okay cool this we can see these Dynamics playing out um which seems to tell us something interesting yes definitely yeah formalizing structurally different ways of thinking about these psychological constructs and and what about like any situations where pessimism or just sort of on the mark accuracy like what is it about situations or Lifestyles or different areas where optimism is relevant and it is it always that way or where wouldn't it be so if there's situation where optimism is not a good thing yeah yeah well I think it really is important to think of an individual's environment so I mean I live in Melbourne I have a great life for me it's like very a good thing to be optimistic because if I go out into my environment there's lots of opportunities so it pays for me to think things are better than they are because I can go out and do all these things and and there's lots of opportunities there I wouldn't say someone in a a serious like if someone's in a war zone I wouldn't say oh you know what you should be optimistic like that's an environment where optimism won't be beneficial and when we're thinking about these interventions that's such an important thing to consider uh yeah so and also if we're if so you could have a a patient who might be in depressed or low optimism because they're in something like a domestic violence situation you wouldn't want to say oh you should um just be optimistic like that will help you want to help fit their environment first so I think the environment is such a

big important part in when optimism is a good thing um but yakob and I have spoken about this a lot if you so this is the idea of the generative model and the generative process if we are optimistic and we have these these positive beliefs and then we engage in these positive actions we do also change we can influence the generative process so we can we can change our environment to be better as well so it's like this it's kind of like a a feedback loop so there are situations where being optimistic because your environment affords opportunities but then that optimism and engagement changes the environment to afford even more opportunities and make things better overall yeah make makes sense some situations [Music] where the agent's beliefs or actions don't change like the slot machine but then there's other situations like maybe interpersonally where getting somebody else excited as part of like a feedback loop that that um okay I'll ask a question from the chat upcycle Club wrote do you think active inference can help computational Psychiatry and even cognitive science in general to model additional cognitive biases beyond the optimism bias um that's a really good question and yeah I think that um we can model I think it lots of different biases with active inference and and other models as well and I think that they're going to be really important in um computational Psychiatry because so much of our behavior and um the things that we do of a lot of it can be from the biases I mean we all have them um all different biases and a lot of them we're not aware of so a lot of them we we think we don't have them but we do and these models are a really good way of of investigating that um I have lots of biases I think I don't have um and and yeah but if I could do a if I could have a model of it and do a task that could look into that without me self-reporting on it I probably would be very helpful um and I guess the the argument for using active inference is when we are going out in so the way that our the active inference modeling works and we're doing actions to kind of fulfill these beliefs that we have it can kind of show how all these different biases could be um sustained I guess not just the optimism one others as well because we're looking for observations and and being in the world to to to fit our model yeah the piece about action as like a self-fulfilling prophecy in active inference is is interesting because in a reward maximization model of behavior selection it's almost like optimism or at least do doing the best of the options that are available that's the nature of action selection itself whereas an act of inference something is being selected because it's believed to be likelier or likeliest so in that sense it sort of seems to on a first pass encode this sort of ball rolling downhill just down the middle things should be just right as they are so then it's interesting to see like what are the ways that you you can model optimism and then the fact that you had it as a second level sits it as a sort of domain General feature or

like cognitive summary variable um so is this like the same generative model that you adapted three ways or or how were across the experiments was it the same graph being used in in slightly different settings yeah it was the same the same model so that was that and and you know I again because we're trying to have a model that has an application at the end I mean ideally if we're thinking about what this is really looking like it would be heaps of levels of models and optimism is that the at the very top but but that's that wasn't our goal is our goal was something that can be used but that yeah I think optimism is definitely this if we're thinking about hierarchical models it would be kind of very much at at the top and shaping our um yeah perception and action on all those things in in lots of different ways um yob want to at add anything or or what what brought you in the lab to want to explore this question other than having an optimistic belief about a cool grad student I mean it's it's just 100% Beth's um initiative to focus on optimism and um the the group as such has a nice kind of active inference Focus um on you know trying to model various things and contemp contemplative practices as well and and which relates to mental health and well-being and optimism is I've discovered through Beth's work here really a key element in this and then there are some nice Beth and I have talked a lot about how how might this connect to the free energy principle and so on and so we've kind of played rather playfully around with ideas of a universal optimism bias um that a self-evidence and agent should have um or might have so trying to so I don't know if you mentioned Beth um that the optimism bias is found in in an astounding range of different creatures um you know M donkeys you knowes birds and so on and so it's not like um you know status bias or things that we we humans have it's not a specifically human bias it's it's really astoundingly Broad and so again Beth and I have talked a lot about well that's a kind of funny thing isn't it that it's it seems more fundamental than other biases or maybe other biases are kind of linked to the optimism bias or something like that and so we've then found some some traces of optimism bias uh in in the work from Thomas par and and so on as well and then we tried to kind of rather speculatively draw a line between the two so so it's a very good example of this this lab this Center is is very led by you know the the projects that uh the PhD students and the postu want to do and I kind of come for the ride which is fantastic and then something magical happens when when these things begin to converge and so on so it's not a coincidence of course that that um Beth's work on optimism here really kind of lifts things up and and kind of gives us this kind of sense of but there's something bigger going on as well and then you know Beth's real Focus as you started talking about Beth is how can this work out there in the world in the clinic and so on um which is also incredibly important

because it's very easy to sit and just be you know super enthusiastic about active inference and the free energy principles what what difference does it make um so that's really nice as well yeah the that's in interesting so so what was the universal optimism bias you I talk about this with yeah yeah yeah so I think one one way to kind of so this is written for a um a kind of celebratory special issue of a journal for Carl Friston 65th birthday some of you might have seen and so it's a little bit tongue and cheek a little bit informal like these puff pieces sometimes are I wouldn't call it puff piece but um you know it's a little bit of a shangra and so on so we allow ourselves to let our hair down a little bit and and and just speculate a bit and there's this nice um very short uh bit in Thomas Parr and Giovanni Pezzulo and Carl Friston's active inference book where and one of Thomas's earlier papers as well where he says um you know if the agent exists it must be able to kind of withstand the dissipative forces of the universe and then Thomas goes on and says this means there's a kind of universal optimism bias that must be in play where in this coupled relation between the agent and the surrounding world that the agent kind of believes that they are have the upper hand and and we interpret that as a universal optimism bias and then we try to draw the line to this behavior that we see in all these different kinds of creatures and organisms out there um to put an idea out there more than to kind of prove anything but um it was kind of an interesting thing to think about because then the optimism bias we can kind of come around in a circle between you know something very fundamental about the free energy principle and self-evidencing and so on and then on the other end you know the focus of Beth's work on on clinical relevance and so on and maybe somehow arrive at something interesting that's cool it's kind of like seeing the first principles grounding for what what is seen in the clinic and then if the beliefs were if if if the uh be or the person or whatever it is believes that things are worse than they are then that could be part of a descending spiral so that the question is how much optimism from sort of like just a tiny bit to delusionally risky and then it it it um sets that as something about differential persistence like you could have whole populations and simulating um Evolution and all this with different kinds of um biases um so what where else slash next will you be heading in the work Beth um so we just have also put out a preprint with some rat data some um it was a study uh giving rats psilocybin which is the ingredient in magic mushrooms and we did active inference model fitting to that data to show that over time um psilocybin was increasing optimism in the rats and uh for that we it's it's kind of the same task Paradigm that the rats can decide to engage in the task or not so so similar to the the two unbanded task I just spoke about um so that's something I'm I'm super excited about because obviously Simon is getting a lot of

interest at the moment with treatments for depression so I think understanding the mechanisms of is super important and so hopefully that work can go to the next step when we're looking to see how those models might work in humans like the cory and and active inference models might work in humans um then data fitting for collecting data for the the studies I spoke about today um we also are coming up with a computational model of how gratitude practice might uh increase optimism and again that that's all thought out with how we can then collect data and and fit and test that as well so lots of different optimism interventions and and modeling how do you see epistemic value in relationship to optimism um like learning and updating um for its own sake rather than expecting utility of a situation well I think that that is inherently optimistic do you do you does that make I just think that that's like that is an optimistic thing to go out there and reduce uncertainty and believe that there's you know just for the sake of like oh well there's going to be something more I think that that that's an optimistic action and belief in itself I think also what we've talked about death often is uh if you don't optimistically engage you might be missing a state transition out there in the environment that might have happened without you knowing so you keep going to parties all parties suck and you keep you optimistically keep going out there and part of it is well maybe you know I go on the you know the 10th party there's been a bit of a change in the social scene uh the toxic person has left I have matured a bit now it will be good so there's been these State Transitions and if I don't engage I won't learn like I won't get the epistemic value about State transitions up so that's a kind of an element of an epistemic side of optimism maybe as well that's very interesting it's sort of like it could be possible if and especially what you brought in in the third example with actions over optimistic state so basically attention or the ability to control one's optimism State I mean that'd be kind of cool but then there'd be a way to weave different foraging bouts so that when you go out and it you go out with high hopes and then when those High Hopes are confirmed it's like great I should be that optimistic but then when it wasn't it could be framed as a learning and then it it could just you could stay in the game and continue playing it's like I'm playing this slot machine because I think it's going to give a high expected reward and then I'm going to play this one because I'm trying to learn what its reward is and then that might keep one doing that engaged action rather than that's very interesting rather than only like putting the thumb on the scale of um utility and pragmatic value yeah exactly um this is kind of related but we found something really cool in the rats where over time so it's like the last days of the was like 14 days after the the psilocybon treatment the rats um who were given the psilocybon had lower loss aversion so that the in the C Matrix like the value of

of the loss was smaller for them so the IDE is is that yeah you can go out somewhere and even if you get something bad it just doesn't affect you as much you're kind of like you're more resilient to that in a way and that's the example of like y I can go to 10 parties and I have a bad time but if I'm optimistic that that's not affecting me and I and I'm yeah I'm still learning I'm still going out there and I'm curious to see like what the 11th party is going to bring which is a good thing it will it will be it it will be a very interesting journey I mean you laid out sort of the road map with going from the the conceptualization of the problem through the simulation and then that will be interesting on the sort of diagnostic sense making side and then on the intervention side and then also so many existing um diagnostic instruments and interventions May kind of fit into this way of um modeling I I'll um ask a question from the chat Courtney asks would you consider a bias the same as an expectation is optimism the expectation of a positive outcome yeah so that's a good question I so like it is an expectation of a good outcome but then it also affects your it's not just the expectation because it also affects your belief updating so it's not just oh I expect the good things to happen when I get the good things they're also changing my beliefs more than um the bad things and I think if it was only an expectation you wouldn't have that asymmetry when you're updating your belief so I think that's why it's more just like that that's probably more the bias component to it it affects different things not just it can be carried out in these different ways not just that that one way whereas if I just have positive expectations that might not necessarily mean when I get the positive thing I update more to that positive outcome as well but they're all good questions like how these different things connect so I would say positive expectations isn't one element of the optimism bias cool um any sort of closing thoughts or what what do you leave people with Slash how could they learn more or or build up their optimism bias or whatever it may be um well thank you so much for inviting us on this has been so nice it's been really cool it's so exciting when you do your own work and you're like how is this fitting to anything or is anyone interested in this at all and then you have nice conversations it's so it's always really positive um yeah I think it's uh I hope that this kind of work gives people hope and I I I hope that there also there's people who want to use active influence in these in these more practical ways it's been helpful and if anyone ever wants to reach out and talk about ways to do that I'm I'm happy CU I to do that cuz I think active inference holds so much promise as a framework and when we can start um getting it out there in the these kind of settings that's that's going to be really cool um yeah and if you can try and be optimistic and if you can't we might have some interventions for you soon awesome yob any last thoughts no I think it's appropriate that

Beth gets last uh word though uh maybe wait until we publish the cying paper before you take medic mushrooms to become mytic we'll do we'll do okay thank you very much both um till the next time and I'm sure it'll be a good one thanks so much Daniel see you bye bye for